

ME 535: Pump Simulation

Fall 2025

Kendall Lutz

Introduction:

In industry, there are many different types of pumps, such as radial piston pumps, external and internal gear pumps, swashplate piston pumps, and more. This project focused specifically on swashplate piston pumps. The aim was to create a simulation to emulate the kinematic features, flows, pressure, leakages, and volumetric efficiency of a 45cc/rev swashplate pump with 11 pistons and a maximum swash plate angle of 18 degrees.

Design Considerations:

Given a swash plate pump with 45cc/rev and 11 pistons, about 4 cc/rev can be calculated for a single piston. A dead volume of 0.1cc was assumed for each piston. Assuming a maximum displacement of 0.035 m (zp_{max}), the radius and diameter of the piston can be calculated using the volume equation for a cylinder shown in equation 1. The value for the displacement is approximated based on documentation for pumps of the same size and industry standards. The cross-sectional area of the piston can then be found using equation 2.

$$\pi r_{pist}^2 h = v \quad (1)$$

$$\pi r_{pist}^2 = A_{pist} \quad (2)$$

Further, knowing that the 11 pistons will be fixed in the piston chamber in a circular formation, the angle between equally spaced pistons was found in equation 3

$$\frac{360 \text{ degrees}}{11 \text{ pistons}} = 32.7 \text{ degrees / piston} \quad (3)$$

Further, the length of the chamber was chosen based on the previously defined zp_{max} and was made to be slightly longer, so that the maximum displacement never exceeds the chamber length. The radius of the piston circle within the chamber was found to be

$$R = \frac{zp_{max}}{2 * \tan(18)} \quad (4)$$

Figure 1 shows this relationship. The formula uses the maximum beta angle and maximum displacement to find the largest possible radius.

The gap between the piston and the piston chamber wall was calculated using equation 5, using general manufacturing tolerances of 0.2% to 0.5% of the diameter of the piston.

$$h_{gap} = 0.004 \times r_{pist} \quad (5)$$

A summary of all the design consideration values is shown in Table 1 below:

Fixed Design Condition	Value
$z_{p_{max}}$	0.0350 m
r_{pist}	0.006 m
h_{gap}	5.08×10^{-5} m
ϕ	32.7 degrees
$l_{chamber}$	0.04 m
R	0.0539 m
V_{dead}	0.1 cc

Table 1: Design Consideration Choices

These design constraints were then used in 3-D modeling software Autodesk Inventor to visualize the pump chamber and verify that the constraints were feasible. The result is shown below.

For the valve plate design, the conditions below were set.

$$A_{orifice} = \begin{cases} 0 & 0 \leq \phi < 70 \\ A_{suc} & 70 \leq \phi < 110 \\ 0 & 110 \leq \phi, < 250 \\ A_{dis} & 250 \leq \phi < 290 \\ 0 & 290 \leq \phi < 360 \end{cases}$$

Initially, simulations were run with the orifice size for suction and discharge being the same as the piston radius. Through this, it was found that the orifice size needed to be much smaller than the piston radius, so an orifice size of 2×10^{-6} m was set. Further, the discharge orifice area and suction orifice area were made to be different, with the discharge area being calculated in equation X. This was done to more closely simulate a real pump, where orifice sizes are not the same.

$$Discharge\ Area = 1.5 * Suction\ Area \quad (6)$$

Seventy-five-degree ramps were also added to reduce large spikes in pressure, simulating pressure relief grooves. A slight overlap between the pressure relief grooves for the suction and discharge was required to reduce large pressure spikes as well.

Plate Valve Design Condition	Value
$A_{suction}$	3×10^{-6} m
$A_{Discharge}$	2×10^{-6} m

Table 2: Plate Valve Design Consideration Choices

Kinematic Modeling:

First, the equation for maximum stroke length was analyzed using the diagram in Figure 1 to find equation 7. This diagram was also used to determine the piston circle radius in the previous design considerations section.

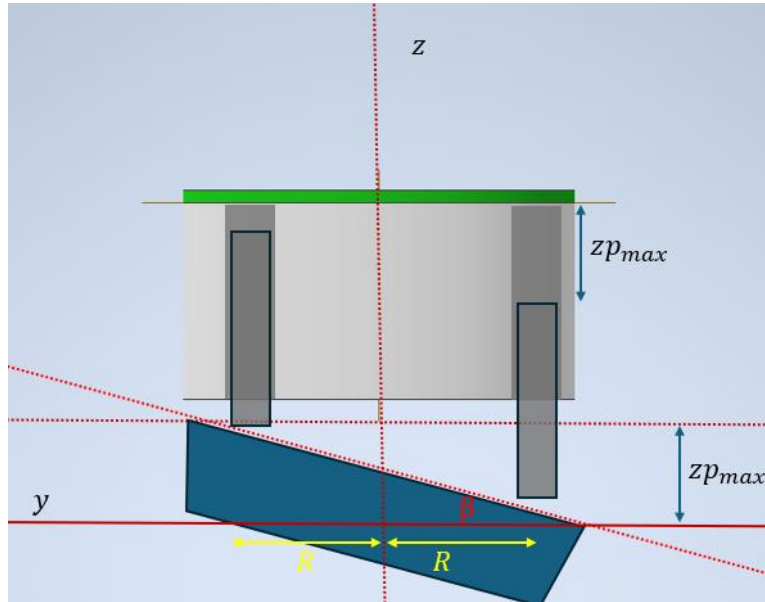


Figure 1: Diagram of swashplate angle relation to stroke length.

$$zp_{max} = 2R \tan(\beta) \quad (7)$$

From this, the equation for max stroke volume could be found and is shown below in equation 8.

$$V_{max} = 2R \tan(\beta) * A_{pist} \quad (8)$$

$$V_{stroke} = 2R \tan(\beta) * A_{pist} \quad (9)$$

This equation can be combined with the equation for instantaneous chamber volume, which gives how the volume changes as the piston moves by angle θ as the shaft rotates.

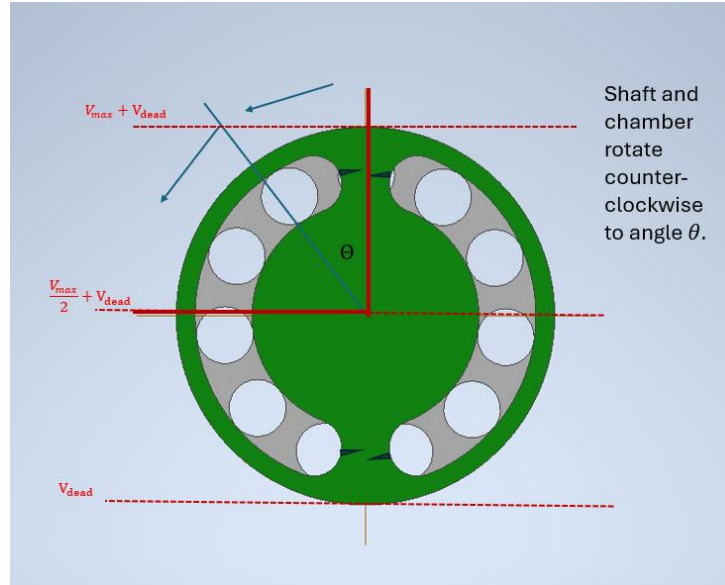


Figure 2: Diagram of angle theta of piston in relation to location on valve plate.

$$V = V_{dead} + \frac{V_{stroke}}{2} + \frac{V_{stroke}}{2} \sin(\omega t) \quad (10)$$

This equation considers the change in angle θ of the piston over time and correlates it to the instantaneous volume when equation 9 for V_{stroke} is used. The combination of equations 9 and 10 assumes that the swashplate path is perfectly sinusoidal.

Fluid Dynamic Modeling:

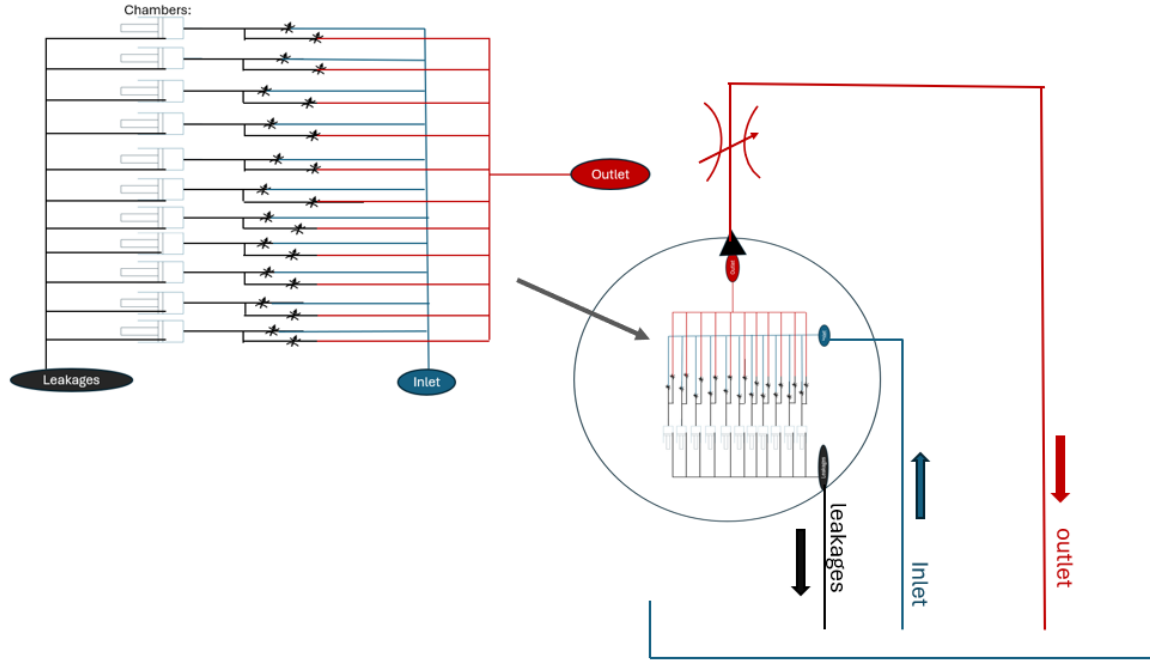


Figure 3: Flow diagram for pump simulation with internal and external components, where red lines are output from the pump, blue is the inlet to the pump, and black is the leakage outlets.

The various subsystems of the simulation pump can be described by examining the path of the flow throughout the entire system. First, fluid from the tank enters the pump through the inlet piping. This flow then enters the pump through the inlet orifice on the plate valve due to the suction pressure from the chamber's increasing volume. This orifice can be modeled using equation 11, and the timing of the inlet and outlet is as described in the design considerations section.

$$Q = C_f A_{orifice} \sqrt{\left(\frac{2\Delta p}{\rho}\right)} \quad (11)$$

As the chamber rotates, the pistons that just received the fluid from the inlet continue to pressurize until they rotate to the outlet orifice, where the flow travels out of the pump. The instantaneous pressure of the chamber can be calculated using equation 12 and the previously defined equation for instantaneous piston chamber volume change.

$$\frac{\delta p}{\delta t} = \frac{K}{V} \left(\sum Q - \frac{\delta V}{\delta t} \right) \quad (12)$$

From the pump, the flow from the pistons can be summed for the total flow rate out of the pump as it travels to the variable orifice, keeping the pump outlet pressure at constant pH, as the flow varies out of the pump. From this orifice, there is a pressure drop, and the flow then travels back to the tank. This relationship between the exit pressure of the pump and the pressure at the inlet

of the orifice is related through equation 13. The area of the orifice changes to maintain a pH of 300 bar.

$$\Delta p = \frac{\left(\frac{fL}{d}\right)pv^2}{2} \quad (13)$$

During this process, Poiseuille and Couette leakages are experienced by each piston chamber as the piston changes in volume. These individual piston leakages are summed for the total leakage flow rate out of the pump. The flow rate of these leakages was determined using equations 14 and 15. The flow from these leakages is redirected back to the tank.

$$Q = \frac{bh^3}{12\mu l} \Delta p \quad (14)$$

$$Q = \frac{bvh}{2} \quad (15)$$

The fluid used in these simulations was ISOVG46 oil, which has a density of $870 \frac{kg}{m^3}$, and a bulk modulus of 17000 bar. The kinematic viscosity of the oil was found to be $4.60 \times 10^{-5} \frac{m^2}{s}$, and the dynamic viscosity was calculated to be 0.0391 Pa-s.

In this simulation, it was assumed there was no leakage from the slipper swashplate interface and no variation in material properties due to pressure. Further, it is assumed that the piston displacement is perfectly sinusoidal and there is no variation in tilt. Moreover, it was assumed that there is a fixed port timing, that is, the piston is open to the inlet and outlet based on the angle θ .

Results:

Below is a table describing the four test conditions for the simulations. RPM and swash plate angle were varied.

Given Conditions			
pH (bar)	pL (bar)	RPM	Swash Plate Angle (%)
300	10	4000	100
300	10	1000	100
300	10	4000	10
300	10	1000	10

Table 3: Test cases for the pump simulation.

Figures 4 – 7 show the flow rate output for the pump at varying swashplate angles and rounds per minute. The flow rate output for the smaller swashplate angle of 10% is always smaller than the output at 100% due to the maximum stroke length decreasing with respect to the angle β . Further, it can be seen that the lower RPM produces a smaller flow rate. This is to be expected as

$Q_{theo} = V_{cc} * RPM / 1000$. As the RPM of the pump increases, the theoretical flow rate increases as well, and this relationship is reflected in the graphs.

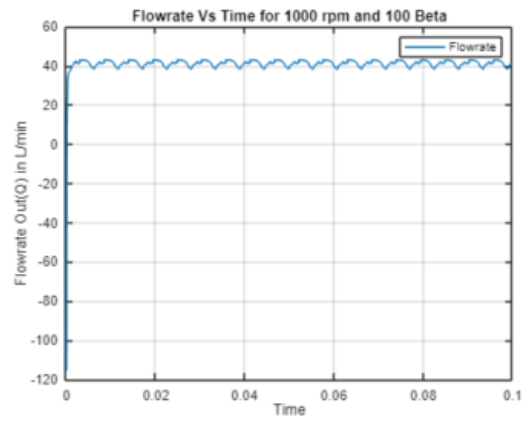


Figure 4: Flow rate vs. time for 1000 RPM and 100% Beta

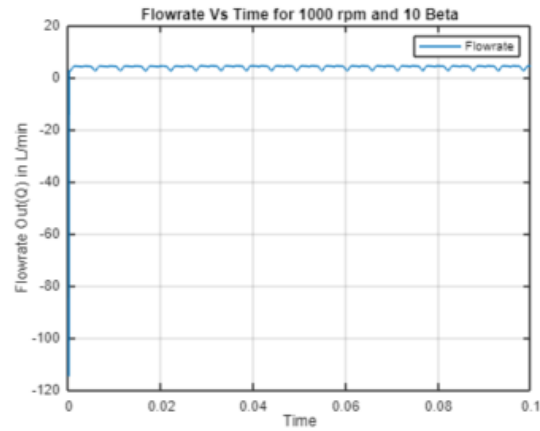


Figure 5: Flow rate vs. time for 1000 RPM and 10% Beta

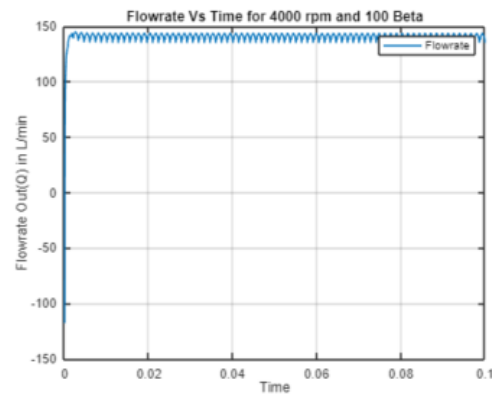


Figure 6: Flow rate vs. time for 4000 RPM and 100% Beta

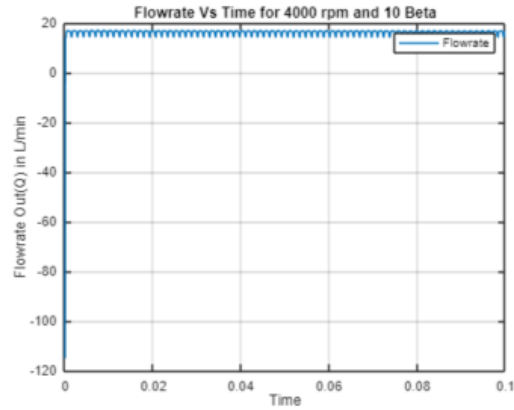


Figure 7: Flow rate vs. time for 4000 RPM and 100% Beta

Next, Figures 8-11 exhibit the pressure immediately after exiting the pump. The pressure for all remains slightly above 300 bar and gradually decreases to 300 bar over the length of the pipe to the orifice. The outlet pressure is higher for greater RPM and beta angles. This logically makes sense as the flow rate out is higher for these parameters; therefore, the line velocity is higher. The resulting pressure drop is found from equation 13.

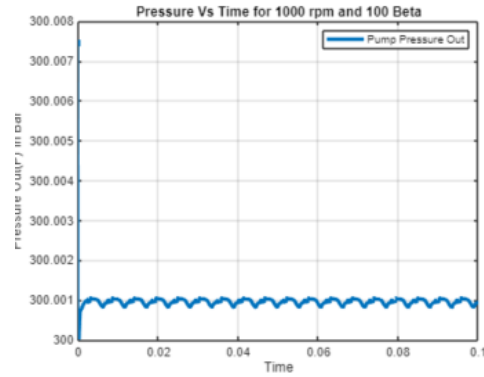


Figure 8: Pressure vs. Time for 1000 RPM and 100% Beta

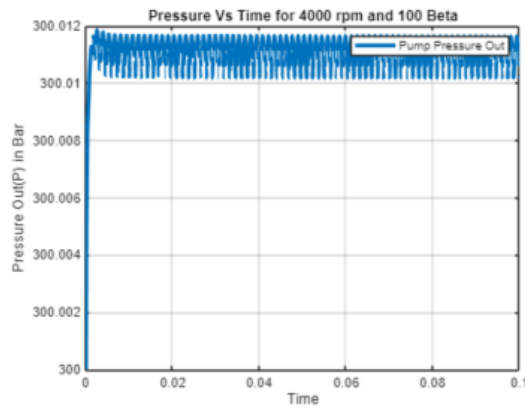


Figure 9: Pressure vs. Time for 1000 RPM and 100% Beta

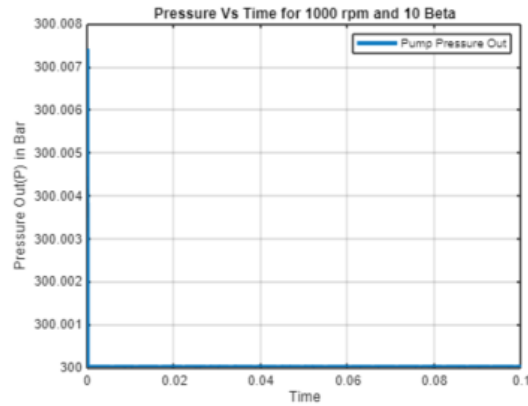


Figure 10: Pressure vs. Time for 4000 RPM and 10% Beta

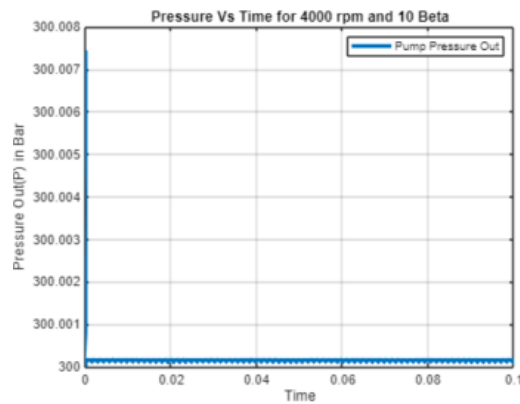


Figure 11: Pressure vs. Time for 4000 RPM and 10% Beta

Next is the volume versus time for the parameters. The minimum volume is the same for all and is equal to the dead volume. The maximum volume is the same for the simulations with the same beta angle. This is because the stroke length is completely dependent on the beta angle and not RPM. However, within the 0.1-second time period, the 4000 RPM has a greater number of peaks, due to the increased speed of the change in volume.

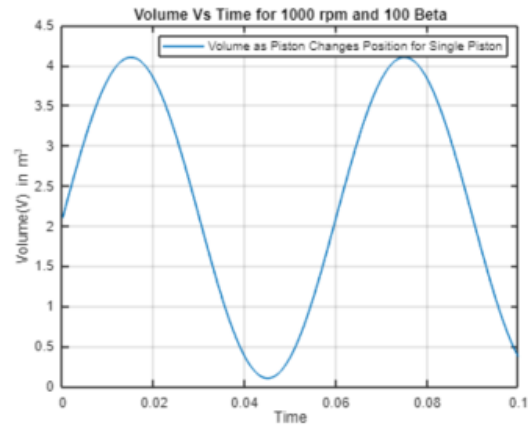


Figure 12: Volume vs. Time for 1000 RPM and 100% Beta

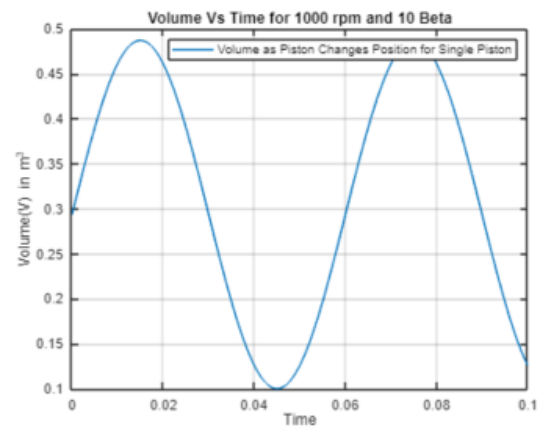


Figure 13: Volume vs. Time for 1000 RPM and 10% Beta

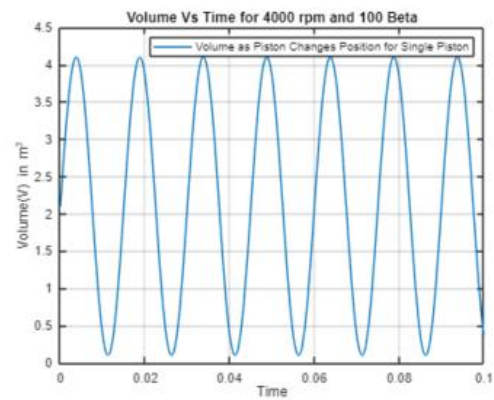


Figure 14: Volume vs. Time for 4000 RPM and 100% Beta

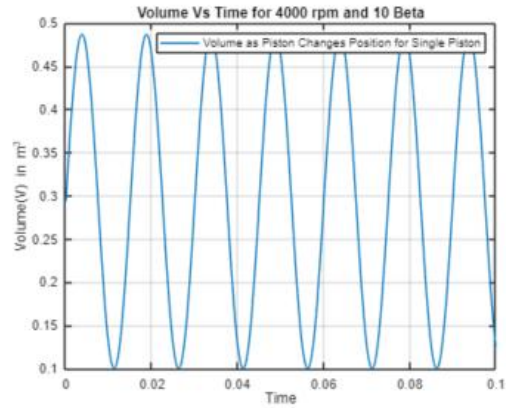


Figure 15: Volume vs. Time for 4000 RPM and 10% Beta

Below are Figures 16 and 17, which show the orifice area over time. There is no change in the maximum orifice areas between the two or the ramp angle. The only change is the increased frequency due to higher RPM. The slight overlap between the lines for the suction and discharge indicates a time when the piston is open to both orifices. This was necessary to reduce extreme pressure spikes. Beta angle does not affect the orifice area.

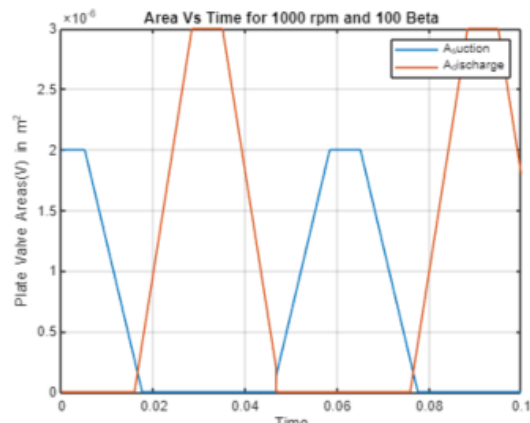


Figure 16: Plate Valve Orifice Area vs. Time for 1000 RPM and 100% Beta

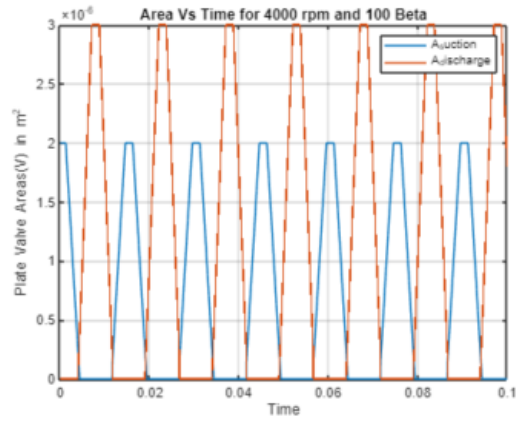


Figure 17: Plate Valve Orifice Area vs. Time for 4000 RPM and 100% Beta

Further, the output parameters for the chamber can be examined. Figures 18-21 show the resultant pressure from the chamber for each case. As expected, utilizing equation 12, due to a larger change in volume from the greater beta angle, the pressure is higher. Further, the greater RPM experiences a higher pressure, due to the derivative time component of equation 12. The time is shorter, so the pressure increases.

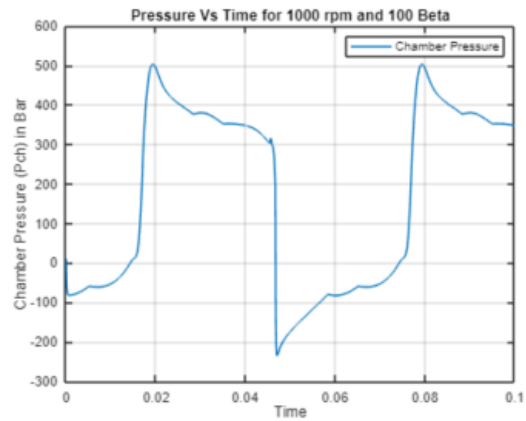


Figure 18: Chamber Pressure vs. Time for 1000 RPM and 100% Beta

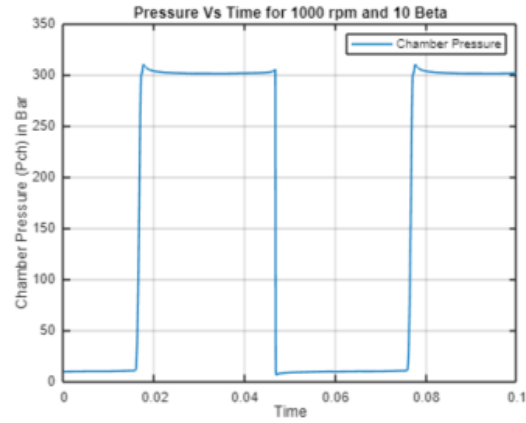


Figure 19: Chamber Pressure vs. Time for 1000 RPM and 10% Beta

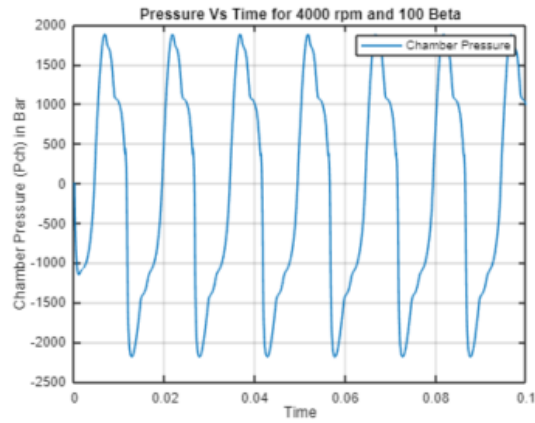


Figure 20: Chamber Pressure vs. Time for 4000 RPM and 100% Beta

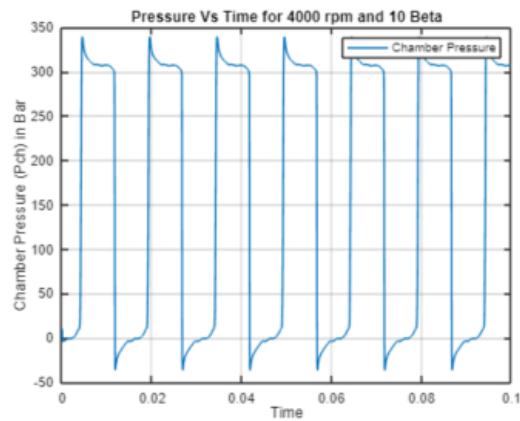


Figure 21: Chamber Pressure vs. Time for 4000 RPM and 100% Beta

The individual chamber flow rates (both inlet and outlet flow rates combined) over time are shown in Figures 22-25. Similar to the overall pump flow rate, the higher the RPM and beta

angle, the greater the flow rate, due to the increased displacement from the beta angle and increased velocity for the higher RPM.

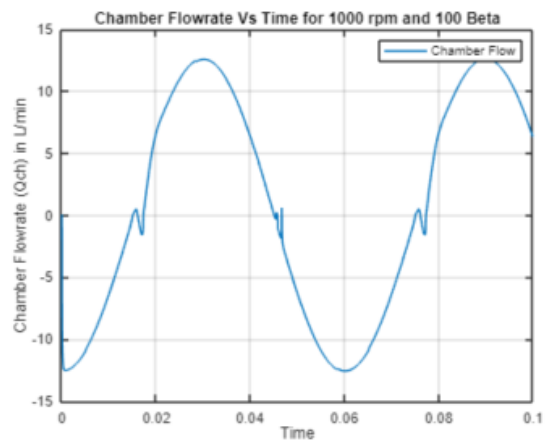


Figure 22: Chamber Flowrate vs. Time for 1000 RPM and 100% Beta

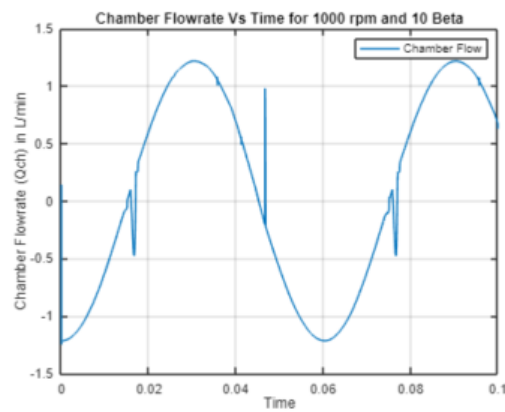


Figure 23: Chamber Flowrate vs. Time for 1000 RPM and 10% Beta

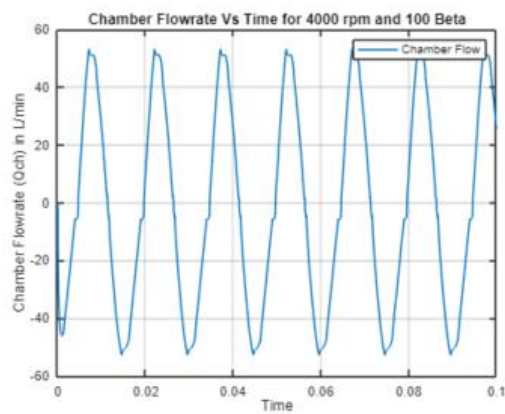


Figure 24: Chamber Flowrate vs. Time for 4000 RPM and 100% Beta

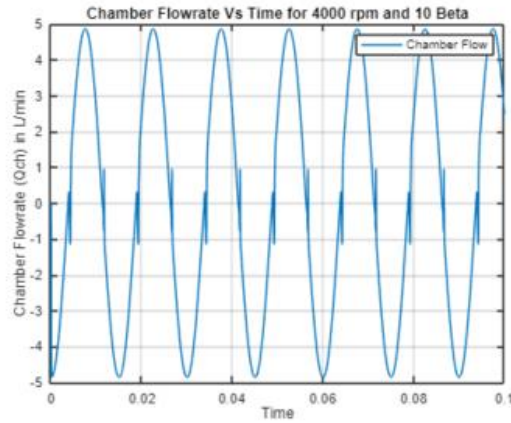


Figure 25: Chamber Flowrate vs. Time for 4000 RPM and 10% Beta

The Figures 26-29 below show the sum of the Poiseuille leakage over time as the piston chamber volume changes. This leakage is dependent on the internal chamber pressure and the stroke length of the piston. Therefore, if there is a higher chamber pressure difference, there should be a higher leakage. This is seen in the graphs below.

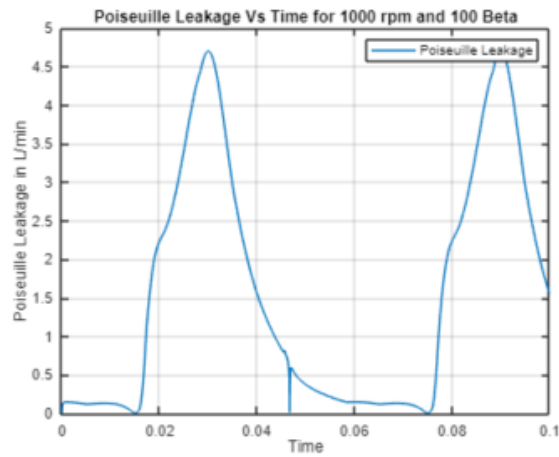


Figure 26: Poiseuille Leakage vs. Time for 1000 RPM and 100% Beta

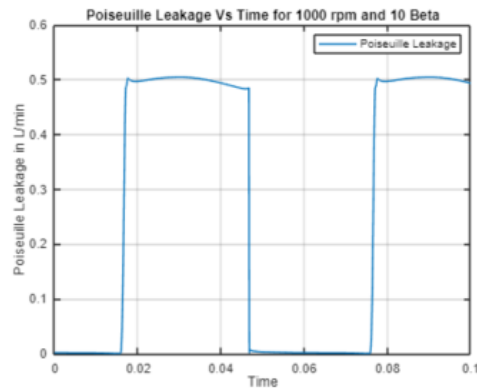


Figure 27: Poiseuille Leakage vs. Time for 1000 RPM and 10% Beta

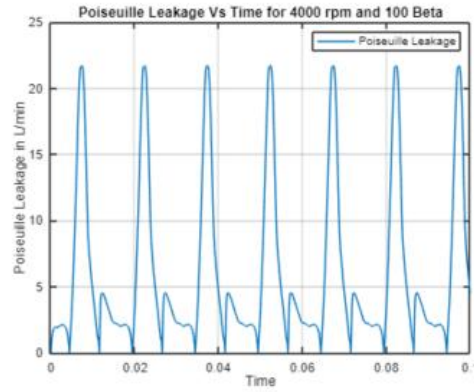


Figure 28: Poiseuille Leakage vs. Time for 4000 RPM and 100% Beta

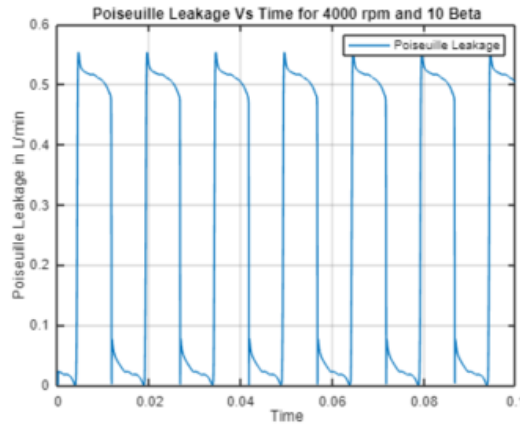


Figure 29: Poiseuille Leakage vs. Time for 4000 RPM and 10% Beta

The Figures 30-33 below show the sum of the Couette leakage over time as the piston chamber volume changes. This leakage is dependent on the velocity and the stroke length of the piston, shown in equation 15. At higher RPM's the velocity is increased, so we expect and see a higher Couette leakage. At higher beta angles percentages the stroke length increases, so we expect to see higher leakages. These effects are seen in the graphs below.

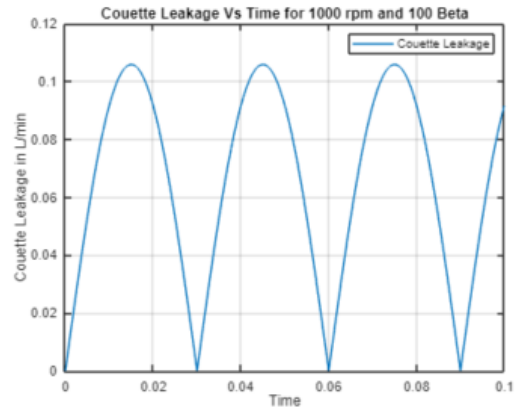


Figure 30: Couette Leakage vs. Time for 1000 RPM and 100% Beta

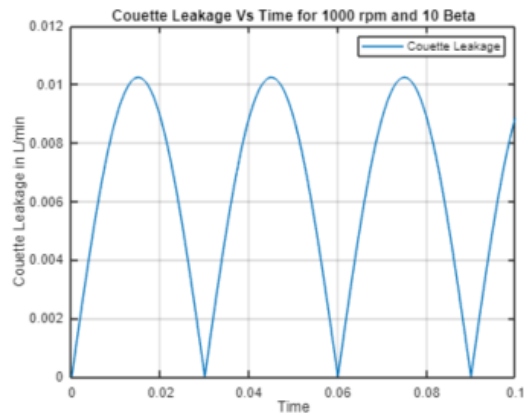


Figure 31: Couette Leakage vs. Time for 1000 RPM and 10% Beta

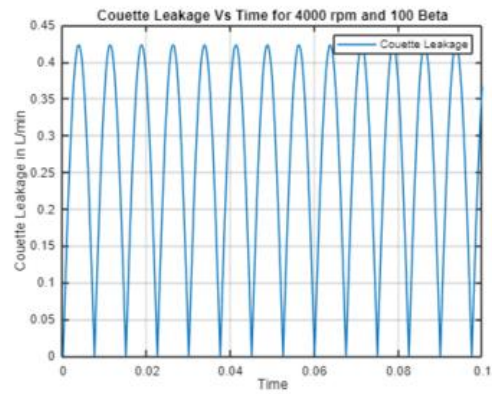


Figure 32: Couette Leakage vs. Time for 4000 RPM and 100% Beta

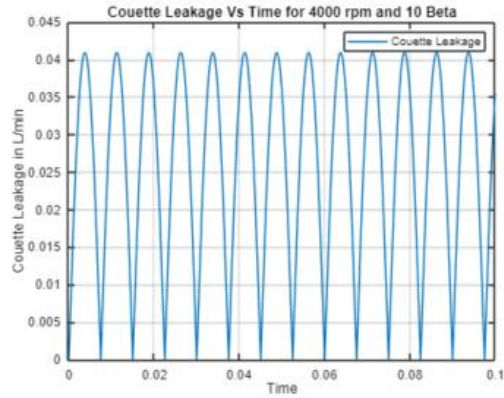


Figure 33: Couette Leakage vs. Time for 4000 RPM and 100% Beta

The variable orifice area to the tank is shown below in Figures 34-37. These reflect the pressure difference between the values for pump outlet pressure in Figures 8-11 and the pH of 300 bar. That pressure difference and the flow rate are used to calculate the area shown in these graphs through the orifice equation. We can expect the Q at 0.02 seconds and Δp at 0.02 seconds input into the orifice equation to give us the resulting area, which are shown in the graphs below.

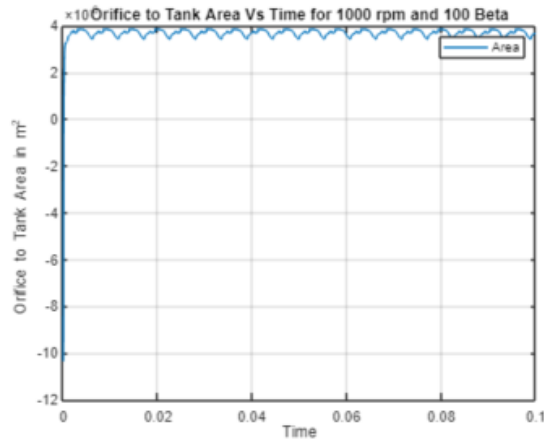


Figure 34: Orifice to Tank Area vs. Time for 1000 RPM and 100% Beta

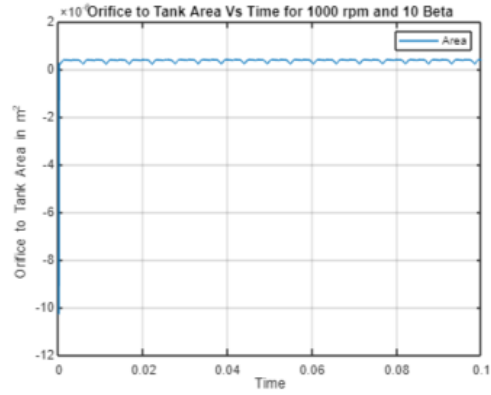


Figure 35: Orifice to Tank Area vs. Time for 1000 RPM and 10% Beta

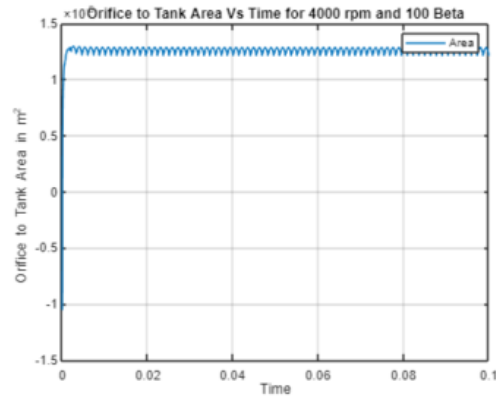


Figure 36: Orifice to Tank Area vs. Time for 4000 RPM and 100% Beta

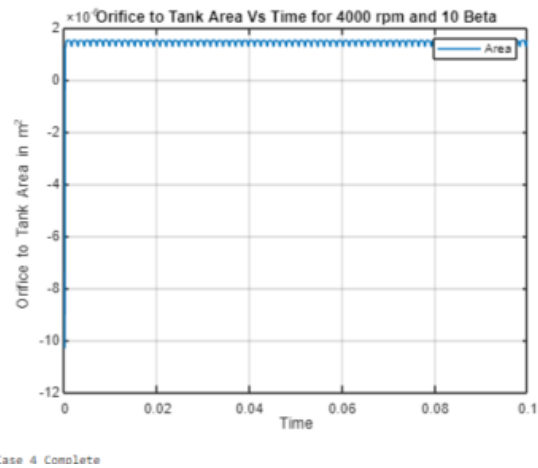


Figure 37: Orifice to Tank Area vs. Time for 4000 RPM and 10% Beta

Finally, the volumetric efficiency for the different pump speeds is shown. At the lower beta angles, the efficiency increases with increasing speed. At higher beta angles, the efficiency decreases with increasing speed. This is due to Couette leakages being greater due to higher

velocity at higher RPMs, and Poiseuille leakages being higher due to the higher pressure. However, at lower beta angles, the magnitude of overall leakages are less, so there is less of a change. The slight increase in efficiency comes from the flow rate increasing faster at higher speeds than the leakage rate.

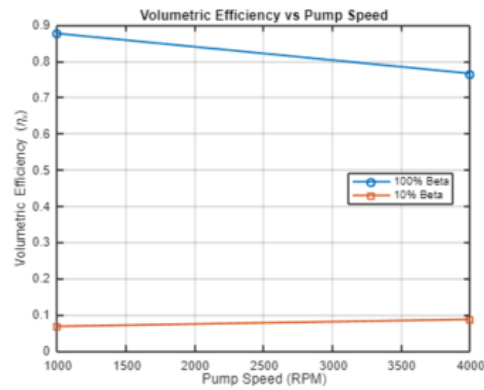


Figure 38: Volumetric Efficiency vs Pump Speed